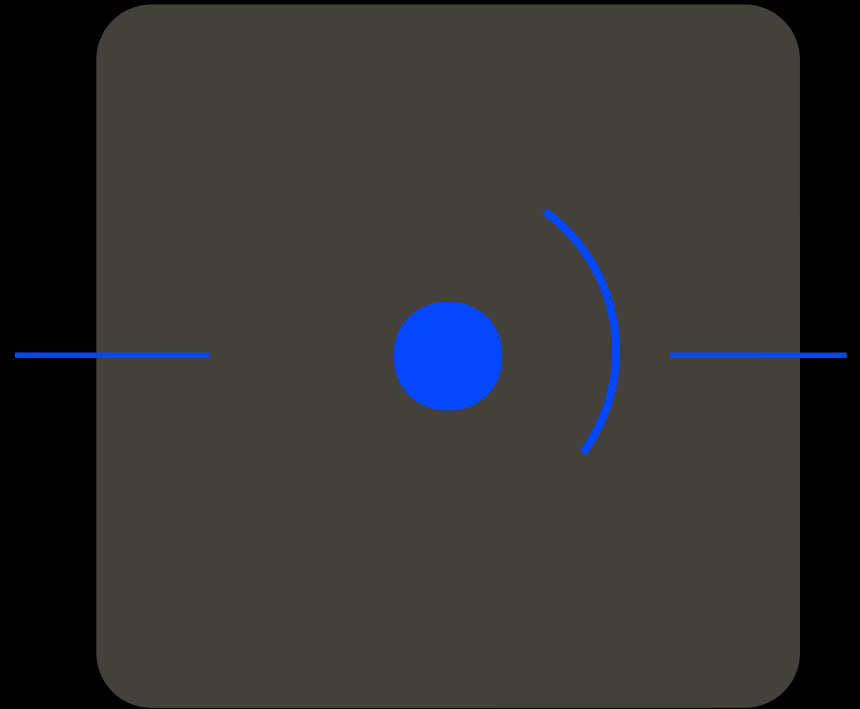


Every alert earns its confidence.

A working single-host pilot for camera onboarding, vehicle and plate analysis, governed face matching, route plausibility, and reviewable evidence.

RUNNABLE PILOT

HONEST EVIDENCE



The camera estate is a decision problem

A useful system must turn heterogeneous streams into defensible operator actions.

80,000+

camera scale in the brief

~160 Gbit/s

at 2 Mbit/s each, before overhead

- **Fragmented sources**

Different departments, protocols, and vendors hide context.

- **Uncertain reads**

Single-frame OCR can turn a weak crop into a confident claim.

- **Alert fatigue**

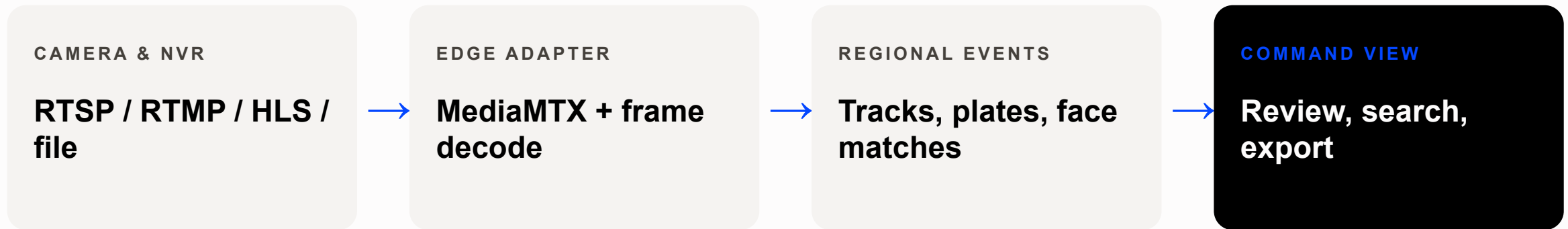
Matches need context, an explicit review path, and provenance.

- **Cost of central video**

Continuous backhaul is the wrong unit of scale; move metadata.

Hybrid by design

One registry and operations view; analysis can move to district or site edge.



Current pilot

Single host: FastAPI + PostgreSQL + detector worker + MediaMTX + React

Reference Models 1–5 require the official portal brief for an accurate side-by-side mapping.

Three mechanisms that reinforce each other

Each keeps weak evidence visible instead of silently upgrading it.

01

Camera Passport

URL or footage onboarding; codec, fps and resolution probed; named department and map point.

02

Route gate

Cross-camera candidates show distance, time gap and implied speed. Impossible jumps enter review.

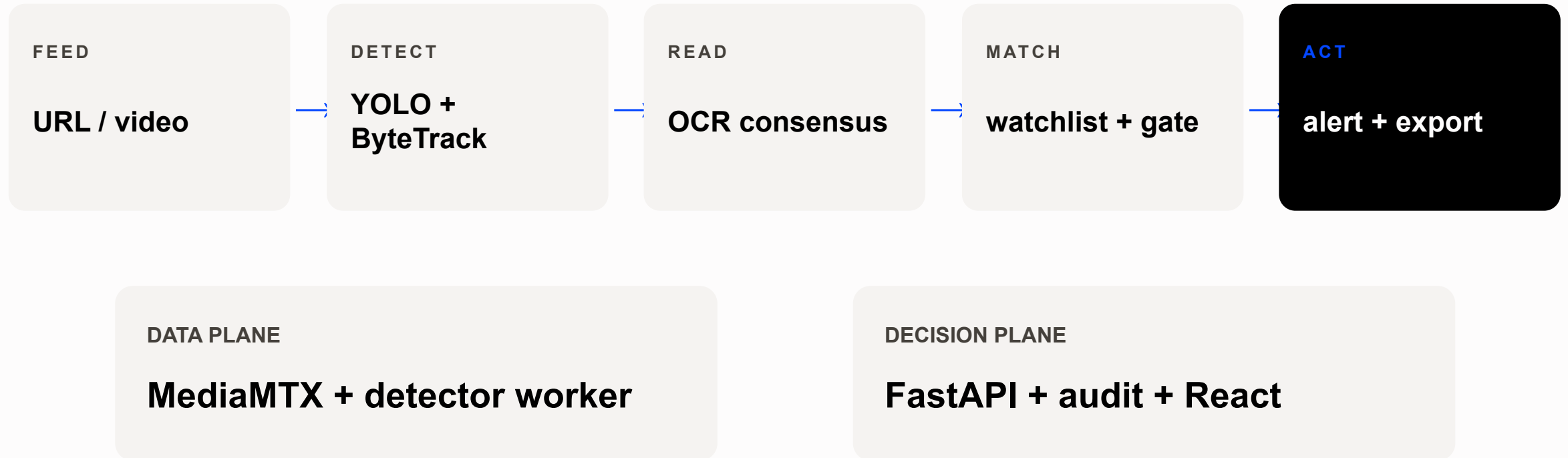
03

Governed FRS

Enrollment needs authority, expiry, admin entry and separate reviewer approval. Unmatched embeddings are discarded.

The working path

A camera becomes a normalized record; a track becomes a reviewable decision.



One vehicle, six accountable steps

The operator can see where the automated chain became uncertain.

01 Onboard

Camera Passport

02 Track

ByteTrack ID

03 Read

Top 3 OCR votes

04 Correlate

Authority + expiry

05 Gate

Distance ÷ time

06 Review

Note + evidence

What the model actually does

No accuracy number is claimed until labelled camera footage is tested.

RUNNABLE ANALYTICS

**Vehicles → tracks → candidate
plate crops → OCR → one
observation per track**

LOW-CONFIDENCE KEPT

- **Optional approved face match**

YuNet + SFace only compare to approved, unexpired entries.

- **Future model work**

Indian plate detector, labelled recall/read-rate study, and the brief's remaining event analytics.

A hit is a structured claim

Identity, authority, expiry, confidence and route context travel together.

MATCH

Exact plate / fuzzy plate / approved face

GATE

Plausible / impossible / insufficient history

TRIAGE

P1 action / human review / reject from action

EVIDENCE

Snapshot hash / model / audit digest

The impossible jump goes to review

One deterministic synthetic run; scenario counts are not field metrics.

ALERT CARD / DEMO

GJ01AB1234

GATE REJECTED

Sarkhej Junction · stolen

Impossible travel in 8 seconds → review queue

09

tracks

07

matched alerts

02

rejected routes

These are scenario counts, not measured field accuracy or false-alert reduction.

Self-hostable core, explicit licensing

The video and decision path can run on a local machine.

Stream	MediaMTX	MIT; localhost-bound pilot
Analytics	YOLO11n + ByteTrack	Ultralytics AGPL / commercial option
OCR + face	Tesseract + OpenCV Zoo	Model and data rights to verify
Data + API	PostgreSQL/PostGIS + FastAPI	Local storage, JWT roles
Console	React + Vite	OpenStreetMap tiles online

Compose defines a reproducible local pilot. Docker engine was unavailable during this verification.

Move events, not every frame

A state deployment is an architecture proposal, not a benchmark result.

160

Gbit/s

80,000 × 2 Mbit/s, continuous
central backhaul

SITE / DISTRICT

Ingest, analytics, short video
retention

REGION

Replayable metadata bus, routing
and storage

STATE

Federated registry, search and
command view

Built for accountable operators

Each decision can be inspected, reversed, and attributed.

Role gates

Operator, administrator and independent face reviewer.

Two-person face approval

Authority and expiry attached before a face entry becomes active.

Audit chain

Each material action includes prior digest; verifier locates tampering.

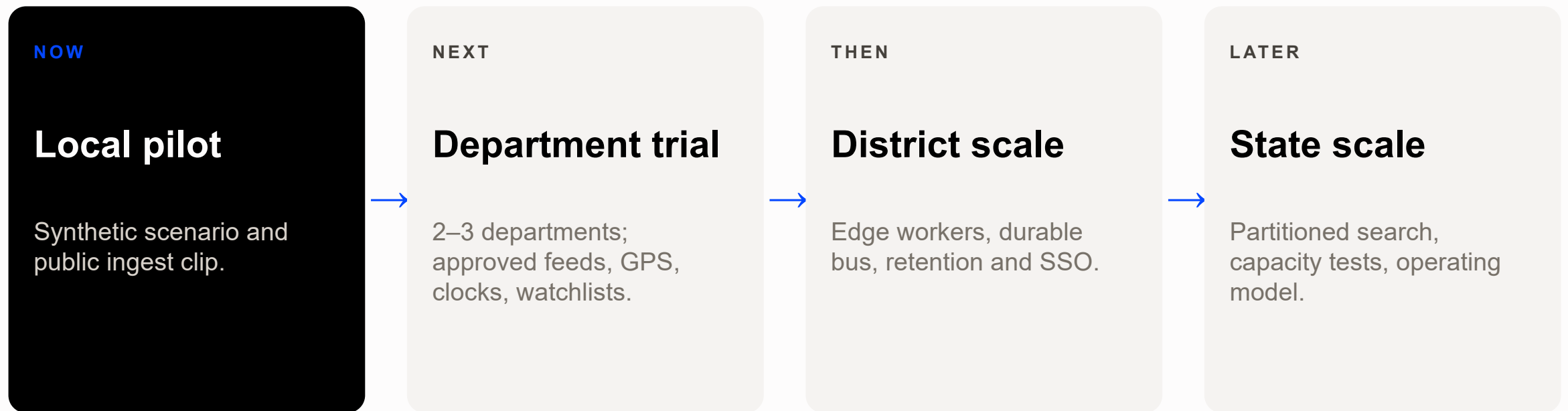
Evidence packet

PDF includes source label, model, snapshot digest, audit digest.

Pilot caveat: localhost network boundary; no external audit anchor or signed forensic seal.

From pilot to department trial

A useful trial starts with authorized sources and labelled evaluation clips.



Prove impact with the next real feed

The product is ready to evaluate; field performance remains unmeasured.

VERIFIED IN THIS BUILD

- API integration tests pass
- Frontend production build passes
- Camera upload and playback checked
- Scenario alerts, route and exports checked
- Face model loading checked

MEASURE ON AUTHORIZED FOOTAGE

- Exact plate read rate
- Vehicle recall
- False alerts before / after gate
- p50 / p95 frame-to-alert latency
- Clock and camera health

Submission status and portal eligibility require organizer confirmation; public notice lists 15 Sep 2026 as the original deadline.